

Experiment 11: Corrected CC18 and CTR23 Results

Tabular Foundation Models, Context Ensemble, and Baselines

July 2026

This document consolidates the four corrected result families. Classification no-shift uses 72 OpenML-CC18 datasets; classification prior shift uses the common 35-dataset binary support with three seeds. Regression uses 35 OpenML-CTR23 datasets. All reported experiment components passed the final coverage audit with no invalid files.

Included result families:

1. Classification: five manipulation families (Accuracy and ROC-AUC).
2. Classification: label-prior shift (Accuracy and ROC-AUC).
3. Regression: five manipulation families (R^2 , NMAE, and NRMSE).
4. Regression: continuous target-prior shift (R^2 , NMAE, and NRMSE).

Table 1: OpenML-CC18 no-shift benchmark at a common manipulation strength of 1 (Accuracy, dataset-level mean).

Group	Model / foundation	Variant	Origin Accuracy	Shortcut Accuracy				Boundary Accuracy				AUC-band Accuracy				Local Accuracy				Mixed Accuracy			
				$\rho = 20$				$\rho = 50$				$\rho = 80$				$\rho = 100$				$\rho = 20$			
				$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$
Traditional ML	LogReg + HPO	-	0.754	0.637	0.461	0.301	0.183	0.661	0.496	0.322	0.215	0.657	0.513	0.377	0.281	0.637	0.463	0.303	0.187	0.643	0.479	0.324	0.216
	SVM + HPO	-	0.777	0.665	0.494	0.332	0.223	0.684	0.528	0.355	0.257	0.679	0.537	0.402	0.310	0.666	0.497	0.338	0.229	0.670	0.509	0.356	0.255
	kNN + HPO	-	0.726	0.640	0.512	0.393	0.307	0.654	0.537	0.415	0.351	0.664	0.570	0.485	0.423	0.642	0.517	0.399	0.315	0.648	0.532	0.424	0.349
	Random Forest + HPO	-	0.776	0.664	0.492	0.331	0.221	0.681	0.521	0.349	0.243	0.687	0.552	0.427	0.339	0.664	0.494	0.335	0.225	0.671	0.509	0.359	0.256
	LightGBM + HPO	-	0.776	0.669	0.504	0.346	0.241	0.687	0.531	0.368	0.268	0.690	0.556	0.429	0.346	0.669	0.507	0.351	0.246	0.676	0.521	0.371	0.275
	CatBoost + HPO	-	0.782	0.673	0.515	0.362	0.255	0.692	0.537	0.379	0.282	0.703	0.586	0.475	0.397	0.674	0.518	0.366	0.259	0.683	0.534	0.394	0.297
	XGBoost + HPO	-	0.774	0.664	0.495	0.335	0.225	0.682	0.522	0.354	0.253	0.689	0.558	0.432	0.346	0.664	0.497	0.338	0.228	0.672	0.513	0.363	0.261
Deep learning	MLP + HPO	-	0.729	0.626	0.469	0.326	0.224	0.649	0.499	0.348	0.268	0.645	0.522	0.407	0.323	0.628	0.474	0.334	0.232	0.633	0.489	0.353	0.260
	FT-Transformer	-	0.745	0.645	0.492	0.343	0.245	0.665	0.522	0.368	0.275	0.667	0.551	0.434	0.360	0.646	0.494	0.352	0.251	0.653	0.509	0.373	0.282
	TabM	-	0.735	0.635	0.485	0.346	0.246	0.656	0.515	0.365	0.272	0.661	0.552	0.446	0.371	0.636	0.488	0.352	0.251	0.642	0.505	0.376	0.284
	TabularRNN	-	0.722	0.626	0.486	0.348	0.254	0.647	0.508	0.367	0.283	0.651	0.551	0.448	0.379	0.627	0.489	0.352	0.259	0.634	0.507	0.375	0.293
	MambaTab	-	0.748	0.640	0.474	0.321	0.213	0.659	0.504	0.342	0.251	0.658	0.522	0.394	0.306	0.641	0.478	0.327	0.222	0.646	0.493	0.348	0.248
	RealMLP	-	0.762	0.652	0.489	0.329	0.222	0.675	0.519	0.354	0.256	0.673	0.543	0.415	0.330	0.654	0.493	0.336	0.231	0.659	0.507	0.360	0.261
	Base		0.804	0.700	0.537	0.380	0.274	0.711	0.558	0.399	0.304	0.717	0.590	0.464	0.378	0.700	0.539	0.384	0.280	0.705	0.553	0.405	0.310
Foundation model	DistPFN		0.801	0.694	0.529	0.365	0.255	0.707	0.549	0.381	0.278	0.714	0.580	0.448	0.353	0.696	0.531	0.369	0.259	0.701	0.545	0.391	0.285
	Context		0.804	0.701	0.602	0.515	0.449	0.712	0.611	0.520	0.463	0.718	0.621	0.547	0.491	0.701	0.605	0.516	0.447	0.706	0.607	0.520	0.462
	Context+DistPFN		0.804	0.701	0.602	0.515	0.449	0.712	0.611	0.520	0.463	0.718	0.621	0.547	0.491	0.701	0.605	0.516	0.447	0.706	0.607	0.520	0.462
	Base		0.782	0.671	0.503	0.341	0.229	0.684	0.523	0.361	0.262	0.685	0.540	0.397	0.300	0.672	0.505	0.346	0.235	0.676	0.517	0.361	0.256
	DistPFN		0.778	0.666	0.499	0.337	0.227	0.679	0.517	0.353	0.256	0.681	0.533	0.388	0.289	0.666	0.502	0.343	0.234	0.670	0.511	0.356	0.249
	Context		0.782	0.677	0.589	0.485	0.412	0.686	0.602	0.498	0.430	0.690	0.608	0.531	0.463	0.677	0.594	0.492	0.409	0.680	0.597	0.494	0.425
	Context+DistPFN		0.782	0.677	0.589	0.485	0.412	0.686	0.602	0.498	0.430	0.690	0.608	0.531	0.463	0.677	0.594	0.492	0.409	0.680	0.597	0.494	0.425
TabPFN-3	Base		0.792	0.677	0.504	0.333	0.220	0.694	0.526	0.356	0.252	0.698	0.559	0.419	0.328	0.678	0.508	0.339	0.227	0.684	0.524	0.361	0.255
	DistPFN		0.788	0.675	0.502	0.331	0.222	0.689	0.523	0.353	0.249	0.695	0.547	0.399	0.302	0.676	0.506	0.338	0.229	0.681	0.517	0.353	0.248
	Context		0.792	0.680	0.587	0.484	0.411	0.695	0.601	0.493	0.428	0.699	0.603	0.528	0.474	0.681	0.591	0.485	0.412	0.684	0.593	0.492	0.428
	Context+DistPFN		0.792	0.680	0.587	0.484	0.411	0.695	0.601	0.493	0.428	0.699	0.603	0.528	0.474	0.681	0.591	0.485	0.412	0.684	0.593	0.492	0.428

Notes. Means use all 72 datasets. Origin is unmanipulated. For each manipulation, ρ is the affected-row percentage. Foundation rows use $\beta = 1$ for classification and severity = 1 for regression; comparison models use the protocol's matching manipulation strength of 1. The manipulation edits query features only. Every method uses the same source-only logistic pseudo-label construction; target labels are used only for scoring. Classification Context uses no additional clean dataset. Only values in Context-containing variants are bold; bold does not denote the column optimum.

Table 2: OpenML-CC18 no-shift benchmark at a common manipulation strength of 1 (ROC-AUC, dataset-level mean).

Group	Model / foundation	Variant	Origin ROC-AUC				Shortcut ROC-AUC				Boundary ROC-AUC				AUC-band ROC-AUC				Local ROC-AUC				Mixed ROC-AUC											
			$\rho = 20$				$\rho = 50$				$\rho = 80$				$\rho = 100$				$\rho = 20$				$\rho = 50$				$\rho = 80$				$\rho = 100$			
			$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$				
Traditional ML	LogReg + HPO	-	0.867	0.739	0.577	0.471	0.409	0.770	0.644	0.536	0.454	0.778	0.658	0.571	0.515	0.742	0.584	0.478	0.416	0.753	0.606	0.499	0.437	0.753	0.606	0.499	0.437	0.753	0.606	0.499	0.437			
	SVM + HPO	-	0.879	0.775	0.626	0.501	0.417	0.804	0.686	0.561	0.468	0.804	0.693	0.598	0.530	0.777	0.632	0.510	0.425	0.786	0.652	0.534	0.453	0.786	0.652	0.534	0.453	0.786	0.652	0.534	0.453			
	kNN + HPO	-	0.856	0.778	0.666	0.557	0.479	0.797	0.703	0.595	0.524	0.804	0.726	0.647	0.587	0.780	0.671	0.564	0.486	0.787	0.689	0.588	0.516	0.787	0.689	0.588	0.516	0.787	0.689	0.588	0.516			
	Random Forest + HPO	-	0.886	0.808	0.683	0.553	0.457	0.827	0.721	0.587	0.481	0.834	0.741	0.643	0.566	0.809	0.686	0.556	0.461	0.816	0.700	0.575	0.482	0.816	0.700	0.575	0.482	0.816	0.700	0.575	0.482			
	LightGBM + HPO	-	0.884	0.797	0.664	0.542	0.457	0.821	0.710	0.581	0.482	0.823	0.721	0.621	0.547	0.798	0.668	0.547	0.462	0.806	0.683	0.567	0.485	0.806	0.683	0.567	0.485	0.806	0.683	0.567	0.485			
	CatBoost + HPO	-	0.890	0.813	0.686	0.555	0.454	0.832	0.728	0.592	0.479	0.837	0.745	0.647	0.571	0.814	0.690	0.559	0.458	0.821	0.705	0.581	0.486	0.821	0.705	0.581	0.486	0.821	0.705	0.581	0.486			
Deep learning	XGBoost + HPO	-	0.885	0.800	0.668	0.541	0.451	0.824	0.714	0.578	0.473	0.827	0.729	0.629	0.553	0.801	0.671	0.545	0.454	0.810	0.687	0.565	0.477	0.810	0.687	0.565	0.477	0.810	0.687	0.565	0.477			
	MLP + HPO	-	0.858	0.754	0.610	0.498	0.427	0.777	0.667	0.554	0.473	0.785	0.681	0.595	0.539	0.757	0.617	0.507	0.435	0.767	0.637	0.531	0.462	0.767	0.637	0.531	0.462	0.767	0.637	0.531	0.462			
	FT-Transformer	-	0.861	0.771	0.639	0.524	0.443	0.794	0.682	0.561	0.471	0.797	0.700	0.607	0.540	0.773	0.644	0.529	0.447	0.779	0.659	0.549	0.471	0.779	0.659	0.549	0.471	0.779	0.659	0.549	0.471			
	TabM	-	0.866	0.767	0.627	0.512	0.435	0.792	0.675	0.556	0.463	0.801	0.701	0.610	0.545	0.768	0.632	0.517	0.438	0.778	0.650	0.541	0.465	0.778	0.650	0.541	0.465	0.778	0.650	0.541	0.465			
	TabularRNN	-	0.856	0.767	0.637	0.531	0.453	0.789	0.681	0.567	0.485	0.798	0.708	0.627	0.567	0.769	0.643	0.536	0.457	0.777	0.660	0.560	0.484	0.777	0.660	0.560	0.484	0.777	0.660	0.560	0.484			
	MambaTab	-	0.869	0.769	0.622	0.495	0.405	0.795	0.672	0.540	0.449	0.800	0.691	0.592	0.517	0.772	0.628	0.502	0.411	0.780	0.647	0.526	0.438	0.780	0.647	0.526	0.438	0.780	0.647	0.526	0.438			
Foundation model	RealMLP	-	0.872	0.774	0.633	0.508	0.422	0.802	0.686	0.556	0.463	0.803	0.700	0.602	0.530	0.776	0.638	0.516	0.429	0.785	0.657	0.537	0.454	0.785	0.657	0.537	0.454	0.785	0.657	0.537	0.454			
	Base		0.901	0.842	0.724	0.578	0.464	0.856	0.761	0.615	0.495	0.857	0.768	0.662	0.576	0.844	0.728	0.585	0.469	0.847	0.739	0.603	0.497	0.847	0.739	0.603	0.497	0.847	0.739	0.603	0.497			
	DistPFN		0.901	0.842	0.722	0.574	0.458	0.855	0.758	0.610	0.489	0.856	0.766	0.656	0.569	0.843	0.725	0.580	0.463	0.847	0.737	0.598	0.491	0.847	0.737	0.598	0.491	0.847	0.737	0.598	0.491			
	Context		0.901	0.843	0.747	0.655	0.593	0.856	0.774	0.679	0.609	0.857	0.773	0.697	0.636	0.844	0.750	0.659	0.589	0.848	0.756	0.665	0.600	0.848	0.756	0.665	0.600	0.848	0.756	0.665	0.600			
	Context+DistPFN		0.901	0.843	0.747	0.655	0.593	0.856	0.774	0.679	0.609	0.857	0.773	0.697	0.636	0.844	0.750	0.659	0.589	0.848	0.756	0.665	0.600	0.848	0.756	0.665	0.600	0.848	0.756	0.665	0.600			
	Base		0.893	0.822	0.687	0.528	0.400	0.837	0.728	0.573	0.445	0.839	0.734	0.609	0.507	0.824	0.691	0.535	0.408	0.829	0.704	0.555	0.437	0.829	0.704	0.555	0.437	0.829	0.704	0.555	0.437			
	DistPFN		0.892	0.821	0.683	0.522	0.395	0.835	0.723	0.565	0.437	0.838	0.730	0.601	0.498	0.822	0.687	0.529	0.402	0.828	0.700	0.548	0.431	0.828	0.700	0.548	0.431	0.828	0.700	0.548	0.431			
	Context		0.893	0.824	0.726	0.629	0.561	0.838	0.757	0.658	0.577	0.841	0.756	0.680	0.616	0.826	0.730	0.635	0.554	0.830	0.737	0.640	0.571	0.830	0.737	0.640	0.571	0.830	0.737	0.640	0.571			
	Context+DistPFN		0.893	0.824	0.726	0.629	0.561	0.838	0.757	0.658	0.577	0.841	0.756	0.680	0.616	0.826	0.730	0.635	0.554	0.830	0.737	0.640	0.571	0.830	0.737	0.640	0.571	0.830	0.737	0.640	0.571			
	Base		0.893	0.819	0.687	0.530	0.408	0.836	0.726	0.571	0.443	0.837	0.737	0.616	0.521	0.821	0.692	0.538	0.415	0.826	0.704	0.557	0.441	0.826	0.704	0.557	0.441	0.826	0.704	0.557	0.441			
TabPFN-3	DistPFN		0.893	0.818	0.683	0.525	0.404	0.835	0.722	0.565	0.438	0.836	0.733	0.609	0.512	0.819	0.687	0.533	0.411	0.824	0.700	0.550	0.435	0.824	0.700	0.550	0.435	0.824	0.700	0.550	0.435			
	Context		0.893	0.820	0.725	0.628	0.562	0.837	0.755	0.657	0.578	0.838	0.750	0.672	0.617	0.822	0.729	0.632	0.559	0.827	0.735	0.640	0.576	0.827	0.735	0.640	0.576	0.827	0.735	0.640	0.576			
	Context+DistPFN		0.893	0.820	0.725	0.628	0.562	0.837	0.755	0.657	0.578	0.838	0.750	0.672	0.617	0.822	0.729	0.632	0.559	0.827	0.735	0.640	0.576	0.827	0.735	0.640	0.576	0.827	0.735	0.640	0.576			
	Context+DistPFN		0.893	0.820	0.725	0.628	0.562	0.837	0.755	0.657	0.578	0.838	0.750	0.672	0.617	0.822	0.729	0.632	0.559	0.827	0.735	0.640	0.576	0.827	0.735	0.640	0.576	0.827	0.735	0.640	0.576			

Notes. Means use all 72 datasets. Origin is unmanipulated. For each manipulation, ρ is the affected-row percentage. Foundation rows use $\beta = 1$ for classification and severity = 1 for regression; comparison models use the protocol's matching manipulation strength of 1. The manipulation edits query features only. Every method uses the same source-only logistic pseudo-label construction; target labels are used only for scoring. Classification Context uses no additional clean dataset. Only values in Context-containing variants are bold; bold does not denote the column optimum.

Table 3: OpenML-CC18 prior-shift curve (Accuracy, dataset-level mean).

Group	Model / foundation	Variant	$s = 0$	$s = 0.25$	$s = 0.50$	$s = 0.75$	$s = 1$
Traditional ML	LogReg + HPO	–	0.765	0.753	0.734	0.706	0.652
	SVM + HPO	–	0.754	0.750	0.713	0.648	0.560
	kNN + HPO	–	0.749	0.734	0.677	0.578	0.467
	Random Forest + HPO	–	0.783	0.781	0.751	0.689	0.570
	LightGBM + HPO	–	0.779	0.775	0.752	0.706	0.607
	CatBoost + HPO	–	0.786	0.782	0.743	0.687	0.586
	XGBoost + HPO	–	0.781	0.781	0.754	0.698	0.590
Deep learning	MLP + HPO	–	0.759	0.751	0.728	0.668	0.574
	FT-Transformer	–	0.757	0.748	0.700	0.628	0.543
	TabM	–	0.766	0.746	0.676	0.561	0.421
	TabulaRNN	–	0.750	0.734	0.675	0.597	0.479
	MambaTab	–	0.771	0.756	0.722	0.678	0.569
	RealMLP	–	0.762	0.767	0.748	0.697	0.606
Foundation model	TabICL-v2	Base	0.788	0.794	0.778	0.729	0.650
		DistPFN	0.775	0.786	0.785	0.761	0.712
		Context	0.789	0.797	0.792	0.786	0.772
		Context+DistPFN	0.788	0.797	0.794	0.788	0.774
	TabDPT	Base	0.779	0.783	0.766	0.712	0.604
		DistPFN	0.764	0.774	0.772	0.750	0.692
		Context	0.780	0.791	0.780	0.770	0.753
		Context+DistPFN	0.781	0.791	0.781	0.769	0.752
	TabPFN-3	Base	0.794	0.799	0.771	0.717	0.619
		DistPFN	0.786	0.797	0.781	0.747	0.683
		Context	0.793	0.802	0.789	0.788	0.760
		Context+DistPFN	0.792	0.802	0.789	0.788	0.765

Notes. Classification uses the common 35-dataset binary CC18 support and seeds 15/16/17. All groups use the identical hard-class source attrition and opposite target-prior construction. Context and DistPFN are source-only corrections and use no target labels; labels are used offline only to instantiate the controlled target prior and to score it. DistPFN is reported only for classification; Context+DistPFN is the frozen combined correction. Only values in Context-containing variants are bold; bold does not denote the column optimum.

Table 4: OpenML-CC18 prior-shift curve (ROC-AUC, dataset-level mean).

Group	Model / foundation	Variant	$s = 0$	$s = 0.25$	$s = 0.50$	$s = 0.75$	$s = 1$
Traditional ML	LogReg + HPO	–	0.819	0.818	0.811	0.806	0.786
	SVM + HPO	–	0.818	0.821	0.814	0.798	0.773
	kNN + HPO	–	0.814	0.817	0.811	0.800	0.769
	Random Forest + HPO	–	0.848	0.849	0.841	0.833	0.810
	LightGBM + HPO	–	0.837	0.841	0.833	0.823	0.794
	CatBoost + HPO	–	0.849	0.850	0.841	0.830	0.808
	XGBoost + HPO	–	0.843	0.845	0.838	0.831	0.803
Deep learning	MLP + HPO	–	0.819	0.817	0.810	0.789	0.710
	FT-Transformer	–	0.816	0.816	0.802	0.794	0.749
	TabM	–	0.823	0.818	0.800	0.790	0.757
	TabulaRNN	–	0.815	0.812	0.795	0.787	0.763
	MambaTab	–	0.831	0.826	0.815	0.810	0.790
	RealMLP	–	0.818	0.822	0.816	0.799	0.757
Foundation model	TabICL-v2	Base	0.863	0.862	0.856	0.846	0.826
		DistPFN	0.863	0.862	0.856	0.846	0.826
		Context	0.863	0.862	0.856	0.846	0.826
		Context+DistPFN	0.863	0.862	0.856	0.846	0.826
	TabDPT	Base	0.855	0.855	0.845	0.840	0.819
		DistPFN	0.855	0.855	0.845	0.840	0.819
		Context	0.855	0.855	0.845	0.840	0.819
		Context+DistPFN	0.855	0.855	0.845	0.840	0.819
	TabPFN-3	Base	0.858	0.862	0.854	0.845	0.819
		DistPFN	0.858	0.862	0.854	0.845	0.819
		Context	0.858	0.862	0.854	0.845	0.819
		Context+DistPFN	0.858	0.862	0.854	0.845	0.819

Notes. Classification uses the common 35-dataset binary CC18 support and seeds 15/16/17. All groups use the identical hard-class source attrition and opposite target-prior construction. Context and DistPFN are source-only corrections and use no target labels; labels are used offline only to instantiate the controlled target prior and to score it. DistPFN is reported only for classification; Context+DistPFN is the frozen combined correction. Only values in Context-containing variants are bold; bold does not denote the column optimum.

Table 5: OpenML-CTR23 no-shift benchmark at a common manipulation strength of 1 (R^2 , dataset-level mean).

Group	Model / foundation	Variant	Origin R^2	Shortcut R^2				Boundary R^2				Rank-band R^2				Local R^2				Mixed R^2			
				$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$
Traditional ML	Ridge + HPO	-	0.543	0.370	0.331	0.358	0.390	0.385	-0.490	-1.464	-3.818	0.388	-0.360	-1.687	-3.528	0.397	0.274	0.016	-0.486	0.367	-0.059	-0.877	-1.673
	SVR + HPO	-	0.492	0.357	0.282	0.280	0.332	0.391	-0.868	-3.416	-3.914	0.331	-0.835	-2.925	-3.614	0.341	0.108	-0.285	-1.019	0.398	0.148	-1.184	-2.344
	kNN + HPO	-	0.587	0.399	0.369	0.382	0.404	0.185	-0.977	-2.226	-3.937	0.174	-0.950	-2.376	-3.640	0.243	-0.948	-2.839	-4.310	0.328	-0.381	-1.710	-3.115
	Random Forest + HPO	-	0.676	0.443	0.224	0.107	-0.147	0.350	-0.745	-2.136	-4.293	0.202	-0.995	-2.517	-3.968	0.615	0.533	0.304	-6.186	0.519	-0.060	-1.890	-4.724
	LightGBM + HPO	-	0.697	0.535	0.367	0.283	0.192	0.271	-0.958	-2.394	-4.672	0.121	-1.129	-2.764	-4.317	0.627	0.532	0.219	-3.931	0.471	-0.324	-2.301	-5.143
Deep learning	CatBoost + HPO	-	0.708	0.504	0.295	0.205	0.070	0.251	-1.023	-2.467	-4.750	0.076	-1.190	-2.858	-4.387	0.667	0.600	0.260	-6.354	0.420	-0.317	-2.255	-5.213
	XGBoost + HPO	-	0.702	0.526	0.362	0.290	0.233	0.211	-1.020	-2.461	-4.731	0.096	-1.166	-2.850	-4.372	0.623	0.498	0.133	-5.465	0.430	-0.398	-2.335	-5.074
	MLP + HPO	-	0.583	0.464	0.322	0.341	0.395	0.143	-0.986	-2.622	-4.826	0.063	-1.211	-2.830	-4.513	0.440	0.234	-0.059	-0.632	0.323	-0.144	-1.407	-3.716
	FT-Transformer	-	0.657	0.466	0.310	0.237	0.245	0.258	-0.950	-2.484	-4.907	0.135	-1.228	-2.957	-4.528	0.491	0.358	-0.051	-3.194	0.384	-0.258	-2.071	-5.387
	TabM	-	0.655	0.439	0.246	0.195	0.355	0.265	-0.835	-2.491	-4.695	0.220	-1.075	-2.788	-4.336	0.437	0.272	-0.018	-0.850	0.362	-0.404	-2.256	-5.049
Foundation model	TabulaRNN	-	0.580	0.360	0.175	0.190	0.208	0.351	-0.590	-1.538	-4.018	0.369	-0.400	-1.908	-3.716	0.279	0.138	-0.052	-1.659	0.364	-0.390	-1.846	-3.331
	MambaTab	-	0.663	0.511	0.386	0.399	0.478	0.341	-0.780	-2.128	-4.568	0.299	-0.852	-2.533	-4.222	0.580	0.414	0.078	-0.657	0.452	-0.122	-1.302	-3.283
	RealMLP	-	0.694	0.572	0.354	0.315	0.397	0.320	-0.983	-2.665	-4.956	0.042	-1.360	-3.181	-4.595	0.625	0.524	0.127	-3.376	0.498	-0.043	-2.273	-5.848
Foundation model	TabICL-v2	Base	0.713	0.570	0.362	0.211	0.253	0.400	-0.941	-2.622	-4.942	0.072	-1.289	-3.059	-4.564	0.697	0.641	0.415	-0.965	0.638	0.012	-1.901	-5.530
	Context	Context	0.713	0.697	0.645	0.608	0.620	0.655	0.318	-0.102	-0.679	0.571	0.229	-0.216	-0.584	0.729	0.714	0.659	0.313	0.714	0.559	0.080	-0.825
	TabDPT	Base	0.786	0.538	0.324	0.209	0.240	0.318	-1.055	-2.769	-5.089	0.094	-1.275	-3.072	-4.700	0.700	0.511	0.228	-1.271	0.636	-0.088	-2.007	-5.460
Foundation model	Context	Context	0.786	0.686	0.633	0.606	0.615	0.628	0.285	-0.144	-0.719	0.571	0.227	-0.224	-0.622	0.725	0.674	0.609	0.241	0.709	0.530	0.053	-0.809
	TabPFN-3	Base	0.731	0.541	0.317	0.154	0.509	0.325	-1.170	-2.997	-5.114	0.022	-1.467	-3.214	-4.720	0.703	0.510	0.457	-2.199	0.665	0.029	-2.333	-6.066
Foundation model	Context	Context	0.731	0.692	0.637	0.595	0.685	0.638	0.266	-0.194	-0.719	0.561	0.187	-0.252	-0.621	0.732	0.683	0.673	0.012	0.722	0.565	-0.024	-0.958
	Notes. Means use all 35 datasets. Origin is unmanipulated. For each manipulation, ρ is the affected-row percentage. Foundation rows use $\beta = 1$ for classification and severity = 1 for regression; comparison models use the protocol's matching manipulation strength of 1. The manipulation corrupts source-context rows. Mixed uses the same severity-1 random seed for every method. Regression Context retains an equal-capacity clean historical anchor and is therefore an extra-information setting. Only values in Context-containing variants are bold; bold does not denote the column optimum.																						

Table 6: OpenML-CTR23 no-shift benchmark at a common manipulation strength of 1 (NMAE, dataset-level mean).

Group	Model / foundation	Variant	Origin NMAE	Shortcut NMAE						Boundary NMAE						Rank-band NMAE						Local NMAE						Mixed NMAE																	
				$\rho = 20$			$\rho = 50$			$\rho = 80$			$\rho = 100$			$\rho = 20$			$\rho = 50$			$\rho = 80$			$\rho = 100$			$\rho = 20$			$\rho = 50$			$\rho = 80$			$\rho = 100$								
				$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$										
Traditional ML	Ridge + HPO	-	0.435	0.502	0.546	0.550	0.535	0.571	0.912	1.214	1.685	0.562	0.868	1.253	1.634	0.531	0.607	0.744	0.944	0.559	0.781	1.058	1.263	0.463	0.513	0.532	0.523	0.474	0.942	1.712	1.749	0.496	0.932	1.551	1.690	0.480	0.603	0.799	1.075	0.458	0.603	1.077	1.376		
	SVR + HPO	-	0.405	0.463	0.513	0.532	0.523	0.474	0.942	1.712	1.749	0.496	0.932	1.551	1.690	0.480	0.603	0.799	1.075	0.458	0.603	1.077	1.376	0.460	0.499	0.507	0.490	0.623	1.033	1.423	1.822	0.613	1.010	1.443	1.762	0.533	0.947	1.501	1.880	0.551	0.868	1.282	1.628		
	kNN + HPO	-	0.383	0.460	0.499	0.507	0.490	0.623	1.033	1.423	1.822	0.613	1.010	1.443	1.762	0.533	0.947	1.501	1.880	0.551	0.868	1.282	1.628	0.330	0.423	0.543	0.625	0.746	0.553	0.954	1.393	1.911	0.554	0.963	1.456	1.846	0.373	0.422	0.541	2.323	0.477	0.790	1.361	1.974	
	Random Forest + HPO	-	0.330	0.423	0.543	0.625	0.746	0.553	0.954	1.393	1.911	0.554	0.963	1.456	1.846	0.373	0.422	0.541	2.323	0.477	0.790	1.361	1.974	0.307	0.377	0.483	0.555	0.577	0.594	1.022	1.449	1.965	0.587	1.004	1.503	1.897	0.353	0.422	0.584	1.839	0.501	0.866	1.443	2.042	
	LightGBM + HPO	-	0.307	0.377	0.483	0.555	0.577	0.594	1.022	1.449	1.965	0.587	1.004	1.503	1.897	0.353	0.422	0.584	1.839	0.501	0.866	1.443	2.042	0.293	0.378	0.503	0.586	0.661	0.586	1.022	1.460	1.981	0.581	1.004	1.512	1.912	0.328	0.379	0.536	2.383	0.496	0.851	1.424	2.051	
	CatBoost + HPO	-	0.293	0.378	0.503	0.586	0.661	0.586	1.022	1.460	1.981	0.581	1.004	1.512	1.912	0.328	0.379	0.536	2.383	0.496	0.851	1.424	2.051	0.300	0.376	0.485	0.556	0.582	0.602	1.025	1.459	1.973	0.582	1.001	1.510	1.904	0.355	0.436	0.615	2.179	0.506	0.873	1.433	2.019	
Deep learning	XGBoost + HPO	-	0.300	0.376	0.485	0.556	0.582	0.602	1.025	1.459	1.973	0.582	1.001	1.510	1.904	0.355	0.436	0.615	2.179	0.506	0.873	1.433	2.019	0.331	0.395	0.484	0.492	0.484	0.591	1.002	1.461	1.962	0.621	1.033	1.507	1.903	0.408	0.528	0.730	0.992	0.530	0.812	1.208	1.722	
	MLP + HPO	-	0.331	0.395	0.484	0.492	0.484	0.591	1.002	1.461	1.962	0.621	1.033	1.507	1.903	0.408	0.528	0.730	0.992	0.530	0.812	1.208	1.722	0.335	0.408	0.500	0.568	0.555	0.588	0.996	1.472	2.036	0.596	1.030	1.559	1.964	0.411	0.483	0.671	1.694	0.536	0.842	1.380	2.079	
	FT-Transformer	-	0.335	0.408	0.500	0.568	0.555	0.588	0.996	1.472	2.036	0.596	1.030	1.559	1.964	0.411	0.483	0.671	1.694	0.536	0.842	1.380	2.079	0.336	0.417	0.529	0.583	0.524	0.585	0.974	1.471	1.982	0.581	1.002	1.522	1.912	0.441	0.540	0.694	1.063	0.544	0.886	1.425	2.017	
	TabM	-	0.336	0.417	0.529	0.583	0.524	0.585	0.974	1.471	1.982	0.581	1.002	1.522	1.912	0.441	0.540	0.694	1.063	0.544	0.886	1.425	2.017	0.399	0.478	0.569	0.583	0.574	0.581	0.941	1.234	1.764	0.568	0.874	1.310	1.708	0.557	0.633	0.712	1.252	0.559	0.895	1.304	1.614	
	TabulaRNN	-	0.399	0.478	0.569	0.583	0.574	0.581	0.941	1.234	1.764	0.568	0.874	1.310	1.708	0.557	0.633	0.712	1.252	0.559	0.895	1.304	1.614	0.334	0.405	0.486	0.503	0.470	0.572	0.969	1.379	1.917	0.584	0.974	1.460	1.852	0.400	0.502	0.694	1.007	0.517	0.809	1.193	1.651	
	MambaTab	-	0.334	0.405	0.486	0.503	0.470	0.572	0.969	1.379	1.917	0.584	0.974	1.460	1.852	0.400	0.502	0.694	1.007	0.517	0.809	1.193	1.651	0.302	0.351	0.472	0.519	0.475	0.563	1.021	1.512	2.074	0.597	1.062	1.615	2.004	0.338	0.391	0.577	1.702	0.478	0.777	1.435	2.208	
Foundation model	RealMLP	-	0.302	0.351	0.472	0.519	0.475	0.563	1.021	1.512	2.074	0.597	1.062	1.615	2.004	0.338	0.391	0.577	1.702	0.478	0.777	1.435	2.208	0.286	0.324	0.447	0.554	0.644	0.496	0.994	1.510	2.055	0.544	1.021	1.581	1.982	0.291	0.340	0.476	1.077	0.394	0.786	1.387	2.139	
	TabICL-v2	Base	0.286	0.324	0.447	0.554	0.644	0.496	0.994	1.510	2.055	0.544	1.021	1.581	1.982	0.291	0.340	0.476	1.077	0.394	0.786	1.387	2.139	Context	0.286	0.278	0.327	0.372	0.420	0.361	0.593	0.837	1.085	0.387	0.609	0.869	1.050	0.270	0.291	0.348	0.615	0.312	0.486	0.765	1.123
	TabDPT	Base	0.242	0.341	0.466	0.560	0.603	0.524	1.010	1.535	2.090	0.556	1.032	1.586	2.015	0.298	0.394	0.547	1.167	0.392	0.837	1.418	2.116	Context	0.242	0.289	0.338	0.378	0.399	0.380	0.605	0.851	1.100	0.397	0.616	0.872	1.065	0.278	0.320	0.382	0.656	0.316	0.450	0.780	1.111
	TabPFN-3	Base	0.276	0.331	0.464	0.584	0.436	0.520	1.050	1.604	2.097	0.554	1.063	1.618	2.022	0.288	0.348	0.451	1.363	0.367	0.793	1.530	2.260	Context	0.276	0.280	0.333	0.385	0.327	0.371	0.618	0.877	1.103	0.390	0.625	0.884	1.066	0.267	0.293	0.333	0.745	0.298	0.485	0.828	1.177
		Context																																											

Notes. Means use all 35 datasets. Origin is unmanipulated. For each manipulation, ρ is the affected-row percentage. Foundation rows use $\beta = 1$ for classification and severity = 1 for regression; comparison models use the protocol's matching manipulation strength of 1. The manipulation corrupts source-context rows. Mixed uses the same severity-1 random seed for every method. Regression Context retains an equal-capacity clean historical anchor and is therefore an extra-information setting. Only values in Context-containing variants are bold; bold does not denote the column optimum.

Table 7: OpenML-CTR23 no-shift benchmark at a common manipulation strength of 1 (NRMSE, dataset-level mean).

Group	Model / foundation	Variant	Origin NRMSE	Shortcut NRMSE						Boundary NRMSE						Rank-band NRMSE						Local NRMSE						Mixed NRMSE											
				$\rho = 20$			$\rho = 50$			$\rho = 80$			$\rho = 100$			$\rho = 20$			$\rho = 50$			$\rho = 80$			$\rho = 100$			$\rho = 20$			$\rho = 50$			$\rho = 80$			$\rho = 100$		
				$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$	$\rho = 20$	$\rho = 50$	$\rho = 80$	$\rho = 100$				
Traditional ML	Ridge + HPO	-	0.684	0.813	0.835	0.821	0.798	0.810	1.177	1.534	2.142	0.823	1.176	1.613	2.081	0.802	0.883	1.020	1.239	0.831	1.072	1.390	1.627																
	SVR + HPO	-	0.703	0.814	0.864	0.866	0.832	0.780	1.268	2.050	2.130	0.838	1.328	1.932	2.069	0.822	0.970	1.155	1.401	0.785	0.950	1.474	1.772																
	kNN + HPO	-	0.656	0.810	0.826	0.811	0.786	0.936	1.381	1.760	2.177	0.951	1.398	1.807	2.114	0.892	1.412	1.925	2.240	0.865	1.210	1.648	1.990																
	Random Forest + HPO	-	0.561	0.773	0.912	0.977	1.082	0.824	1.272	1.718	2.250	0.922	1.399	1.835	2.184	0.629	0.696	0.841	2.577	0.728	1.067	1.692	2.326																
	LightGBM + HPO	-	0.529	0.697	0.815	0.874	0.906	0.873	1.359	1.791	2.328	0.968	1.447	1.895	2.258	0.608	0.695	0.880	2.175	0.765	1.176	1.805	2.415																
	CatBoost + HPO	-	0.514	0.727	0.868	0.930	0.986	0.882	1.374	1.808	2.341	0.987	1.460	1.915	2.269	0.571	0.641	0.832	2.621	0.789	1.168	1.786	2.425																
Deep learning	XGBoost + HPO	-	0.525	0.707	0.828	0.876	0.864	0.896	1.367	1.805	2.336	0.973	1.451	1.911	2.265	0.617	0.722	0.936	2.456	0.786	1.195	1.803	2.395																
	MLP + HPO	-	0.568	0.709	0.795	0.777	0.743	0.875	1.337	1.825	2.358	0.969	1.457	1.900	2.296	0.681	0.840	1.048	1.302	0.802	1.106	1.551	2.101																
	FT-Transformer	-	0.568	0.750	0.844	0.902	0.868	0.858	1.322	1.806	2.371	0.961	1.455	1.934	2.298	0.706	0.799	1.016	2.039	0.815	1.140	1.739	2.453																
	TabM	-	0.577	0.768	0.896	0.931	0.820	0.867	1.303	1.810	2.324	0.922	1.420	1.898	2.254	0.760	0.878	1.020	1.383	0.831	1.211	1.792	2.387																
	TabulaRNN	-	0.648	0.826	0.937	0.937	0.903	0.825	1.217	1.555	2.180	0.831	1.193	1.670	2.118	0.863	0.964	1.052	1.599	0.834	1.215	1.677	2.019																
	MambaTab	-	0.564	0.710	0.788	0.782	0.727	0.828	1.281	1.718	2.296	0.878	1.356	1.836	2.228	0.659	0.790	1.002	1.311	0.773	1.093	1.530	2.024																
Foundation model	RealMLP	-	0.527	0.648	0.791	0.821	0.751	0.837	1.363	1.850	2.388	1.005	1.516	1.992	2.318	0.593	0.677	0.902	2.012	0.737	1.057	1.794	2.550																
	TabCL-v2	Base	0.477	0.649	0.798	0.900	0.831	0.779	1.351	1.848	2.371	0.976	1.493	1.966	2.298	0.533	0.593	0.767	1.417	0.617	1.034	1.699	2.478																
	Context	Context	0.477	0.535	0.589	0.632	0.596	0.594	0.844	1.072	1.317	0.677	0.909	1.128	1.283	0.495	0.517	0.585	0.869	0.525	0.698	1.001	1.369																
	TabDPT	Base	0.413	0.687	0.831	0.904	0.860	0.839	1.383	1.878	2.408	0.974	1.491	1.966	2.333	0.538	0.677	0.856	1.489	0.623	1.080	1.724	2.457																
	Context	Context	0.413	0.555	0.608	0.638	0.616	0.627	0.863	1.089	1.335	0.681	0.911	1.131	1.300	0.504	0.559	0.626	0.900	0.533	0.722	1.014	1.360																
	TabPFN-3	Base	0.483	0.670	0.836	0.944	0.684	0.833	1.429	1.944	2.407	1.003	1.550	2.004	2.332	0.534	0.615	0.728	1.727	0.589	1.024	1.813	2.581																
Context	Context	0.483	0.542	0.606	0.652	0.550	0.616	0.876	1.114	1.332	0.687	0.932	1.143	1.297	0.495	0.527	0.564	1.011	0.512	0.692	1.052	1.416																	

Notes. Means use all 35 datasets. Origin is unmanipulated. For each manipulation, ρ is the affected-row percentage. Foundation rows use $\beta = 1$ for classification and severity = 1 for regression; comparison models use the protocol's matching manipulation strength of 1. The manipulation corrupts source-context rows. Mixed uses the same severity-1 random seed for every method. Regression Context retains an equal-capacity clean historical anchor and is therefore an extra-information setting. Only values in Context-containing variants are bold; bold does not denote the column optimum.

Table 8: OpenML-CTR23 prior-shift curve (R^2 , dataset-level mean).

Group	Model / foundation	Variant	$s = 0$	$s = 0.25$	$s = 0.50$	$s = 0.75$	$s = 1$
Traditional ML	Ridge + HPO	–	0.543	0.449	0.200	-0.239	-0.689
	SVR + HPO	–	0.492	0.394	0.199	-0.055	-0.330
	kNN + HPO	–	0.587	0.482	0.196	-0.285	-0.762
	Random Forest + HPO	–	0.676	0.624	0.469	0.177	-0.185
	LightGBM + HPO	–	0.697	0.645	0.503	0.253	-0.097
	CatBoost + HPO	–	0.708	0.661	0.544	0.339	0.026
	XGBoost + HPO	–	0.702	0.654	0.521	0.233	-0.067
Deep learning	MLP + HPO	–	0.582	0.515	0.378	0.054	-0.216
	FT-Transformer	–	0.657	0.574	0.441	0.222	-0.179
	TabM	–	0.655	0.582	0.438	0.246	-0.066
	TabulaRNN	–	0.580	0.520	0.336	-0.191	-0.345
	MambaTab	–	0.663	0.605	0.479	0.274	-0.020
	RealMLP	–	0.694	0.629	0.540	0.317	0.042
Foundation model	TabICL-v2	Base	0.739	0.697	0.641	0.504	0.311
		Context	0.739	0.727	0.715	0.681	0.632
	TabDPT	Base	0.731	0.698	0.616	0.456	0.243
		Context	0.731	0.724	0.702	0.661	0.609
	TabPFN-3	Base	0.739	0.707	0.630	0.473	0.283
		Context	0.739	0.734	0.715	0.675	0.628

Notes. Regression uses all 35 CTR23 datasets; s maps to $\beta = 4s$. All groups use identical source/target rows, hard direction, and effective sampling fractions. Context uses an equal-capacity clean historical anchor and is an extra-information setting. DistPFN is not defined for the continuous-target regression track. Only values in Context-containing variants are bold; bold does not denote the column optimum.

Table 9: OpenML-CTR23 prior-shift curve (NMAE, dataset-level mean).

Group	Model / foundation	Variant	$s = 0$	$s = 0.25$	$s = 0.50$	$s = 0.75$	$s = 1$
Traditional ML	Ridge + HPO	–	0.435	0.490	0.611	0.750	0.889
	SVR + HPO	–	0.405	0.458	0.556	0.671	0.779
	kNN + HPO	–	0.383	0.455	0.596	0.766	0.917
	Random Forest + HPO	–	0.330	0.374	0.465	0.603	0.741
	LightGBM + HPO	–	0.307	0.350	0.440	0.564	0.704
	CatBoost + HPO	–	0.293	0.335	0.414	0.525	0.663
	XGBoost + HPO	–	0.300	0.338	0.415	0.540	0.659
Deep learning	MLP + HPO	–	0.332	0.375	0.446	0.538	0.621
	FT-Transformer	–	0.335	0.382	0.462	0.549	0.678
	TabM	–	0.336	0.377	0.457	0.560	0.681
	TabulaRNN	–	0.399	0.451	0.549	0.664	0.813
	MambaTab	–	0.334	0.374	0.442	0.546	0.664
	RealMLP	–	0.302	0.339	0.391	0.491	0.622
Foundation model	TabICL-v2	Base	0.257	0.289	0.331	0.406	0.504
		Context	0.257	0.269	0.284	0.315	0.356
	TabDPT	Base	0.266	0.294	0.345	0.428	0.534
		Context	0.266	0.276	0.296	0.328	0.372
	TabPFN-3	Base	0.255	0.286	0.340	0.415	0.512
		Context	0.255	0.265	0.286	0.316	0.355

Notes. Regression uses all 35 CTR23 datasets; s maps to $\beta = 4s$. All groups use identical source/target rows, hard direction, and effective sampling fractions. Context uses an equal-capacity clean historical anchor and is an extra-information setting. DistPFN is not defined for the continuous-target regression track. Only values in Context-containing variants are bold; bold does not denote the column optimum.

Table 10: OpenML-CTR23 prior-shift curve (NRMSE, dataset-level mean).

Group	Model / foundation	Variant	$s = 0$	$s = 0.25$	$s = 0.50$	$s = 0.75$	$s = 1$
Traditional ML	Ridge + HPO	–	0.684	0.743	0.873	1.047	1.223
	SVR + HPO	–	0.703	0.760	0.873	1.004	1.122
	kNN + HPO	–	0.656	0.726	0.893	1.101	1.281
	Random Forest + HPO	–	0.561	0.600	0.711	0.888	1.063
	LightGBM + HPO	–	0.529	0.573	0.685	0.842	1.014
	CatBoost + HPO	–	0.514	0.555	0.650	0.800	0.981
	XGBoost + HPO	–	0.525	0.562	0.653	0.828	0.990
Deep learning	MLP + HPO	–	0.569	0.606	0.700	0.849	0.999
	FT-Transformer	–	0.568	0.625	0.733	0.863	1.038
	TabM	–	0.577	0.626	0.736	0.860	1.021
	TabulaRNN	–	0.648	0.682	0.792	0.998	1.135
	MambaTab	–	0.564	0.602	0.694	0.823	0.989
	RealMLP	–	0.527	0.580	0.640	0.804	0.965
Foundation model	TabICL-v2	Base	0.476	0.506	0.563	0.684	0.821
		Context	0.476	0.482	0.499	0.545	0.599
	TabDPT	Base	0.488	0.512	0.589	0.716	0.856
		Context	0.488	0.492	0.517	0.563	0.618
	TabPFN-3	Base	0.476	0.503	0.567	0.682	0.823
		Context	0.476	0.479	0.498	0.541	0.597

Notes. Regression uses all 35 CTR23 datasets; s maps to $\beta = 4s$. All groups use identical source/target rows, hard direction, and effective sampling fractions. Context uses an equal-capacity clean historical anchor and is an extra-information setting. DistPFN is not defined for the continuous-target regression track. Only values in Context-containing variants are bold; bold does not denote the column optimum.